

MCV4UY - Unit 4 Test - More Derivatives

Name _____

Date _____

Total
/36

KNOWLEDGE

1. Differentiate explicitly. Express final answers with positive exponents.

a. $y = \frac{(6x^2-3)^6}{\sqrt[3]{5x+2}}$

b. $y = \sin^{-1}(\cos x)$

(2)

(2)

c. $y = e^{\tan\left(\frac{x^2}{3x-2}\right)}$

d. $y = \log(3 \cos x) + \ln(\tan 3x)$

(2)

(3)

2. Find the second derivative by implicit differentiation. Simply as much as possible.

$$x^3 + y^3 = 1$$

(3)

3. Find $\frac{dy}{dx}$ at $x = 1$ if $y = \frac{1+u}{1+u^2}$ and $u = x^2 - 1$.

(2)

APPLICATION

4. Given the parametric equations, $x = 2 \sin(1 + 3t)$ and $y = 2t^3$,

a. Differentiate

(2)

b. find the equation of the tangent at $t = -\frac{1}{3}$

(2)

5. The function, $P(t) = \frac{50}{1+8e^{-3t}}$ represents a colony of bacteria in a decomposing food sample. The equation represents a carrying capacity of 50 000, population growth rate of 3 and initial population of 42 000. P is the population in thousands and t is time in minutes.

a. Based on the above details, re-write an equation which represents a carrying capacity of 75 000, growth rate of 2 and initial population of 15 000.

(1)

b. Describe what occurs to the rate when population approaches the carrying capacity.

(1)

c. Determine the rate of growth at 3 minutes with the equation in (a).

(2)

6. Use the information contained in the table with x-values of 1, 3, 5, and 7 to answer the following questions.

a. $\frac{d}{dx} f(g(g(x)))$ when $x = 1$

x	1	3	5	7
$f(x)$	5	1	7	3
$g(x)$	3	7	1	5
$f'(x)$	3	5	7	1
$g'(x)$	5	1	3	7

(2)

b. $\frac{d}{dx} \frac{f(x)+g(x)}{f(g(x))}$ when $x = 3$

(3)

COMMUNICATION

7. Explain why the derivative of $f(x) = \sin^{-1}(x)$ is always positive. Use a diagram to support your answer.

(2)

THINKING

8. Find the equation of the line tangent to the graph $y = \ln(x^2 + 4) - x \arctan\left(\frac{x}{2}\right)$ at $x = 2$. Express your final answer in exact values.

(3)

9. Find all points (x,y) on the graph $x^{2/3} + y^{2/3} = 8$ (see diagram) where the lines tangent to the graph at (x,y) have slope -1.

(4)

